

Accelerated Salt-Spray Corrosion of Welded 316L Stainless Steel Joints

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Executive summary

We exposed 48 butt-welded 316L coupons, produced with three filler metals and two shielding gases, to a neutral salt spray for 1,000 hours. Pitting initiated in the heat-affected zone of every coupon, but its density varied fivefold with the welding parameters. Coupons welded with argon-2% nitrogen shielding gas and a 316LSi filler showed the fewest pits (3.1 per cm²) and the shallowest maximum pit depth (42 µm). Post-weld pickling reduced pit density by a further 60%. We recommend this combination for the marine handrail project and propose a field trial.

1. Background

Weld zones in austenitic stainless steels are prone to localised corrosion because chromium is depleted near the fusion line and because heat tint oxides are less protective than the passive film. Earlier work in this laboratory (TR-2024-11) examined base metal only.

2. Test method

- Coupons: 100 × 50 × 3 mm, single-pass TIG butt welds, eight per parameter set.
- Exposure: 5% NaCl fog at 35 °C, continuous, 1,000 hours; coupons inclined at 20°.
- Evaluation: optical counting of pits at 10× magnification; pit depth by focus-variation microscopy; mass loss after cleaning.
- Statistics: two-way ANOVA on log-transformed pit density.

3. Results

Filler / shielding gas	Pit density (per cm ²)	Max pit depth (µm)	Mass loss (g m ⁻²)
316L / argon	11.4	118	6.2
316LSi / argon	8.7	96	5.1
316L / argon + 2% N ₂	4.9	61	3.3
316LSi / argon + 2% N ₂	3.1	42	2.4

Both the shielding gas ($p < 0.001$) and the filler ($p = 0.02$) had significant effects, with no significant interaction. Pits clustered 1–3 mm from the fusion line, coinciding with the darkest heat tint.

4. Conclusions and recommendations

1. Use argon-2% nitrogen shielding gas with 316LSi filler for marine-exposed welds.
2. Specify post-weld pickling and passivation for all visible welds.
3. Run a 12-month field exposure at the harbour test rack before final approval.

References

TR-2024-11, Pitting resistance of 316L sheet in chloride fog, Kestrel Materials Laboratory, 2024.